

JEE ADVANCED - 7 | PAPER - 1

JEE 2022

Date	Maximam Marks	Duration
30th January, 2022	180	3 Hours

General Instructions

- The question paper consists of 3 Subject (Subject I: **Physics**, Subject II: **Chemistry**, Subject III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
- Section 1** contains **2 types** of questions [**Type A** and **Type B**].
Type A contains **SIX (06) Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
Type B contains **SIX (06) Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.
- Section 2** contains **THREE (03) Paragraphs**. There are **TWO (02)** questions corresponding to each Paragraph. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code**, **Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

MARKING SCHEME

SECTION-1 | Type A

- This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the answer. For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

SECTION-1 | Type B

- This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is(are) chosen;
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
Zero Marks : 0 If unanswered;
Negative Marks : -2 In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then
choosing **ONLY** (A), (B) and (D) will get +4 marks; choosing **ONLY** (A) and (D) will get +2 marks;
choosing **ONLY** (A) will get +1 mark;
choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
choosing any other option(s) will get -2 marks.

SECTION-2

- This section contains **THREE (03)** Paragraphs. There are **TWO (02)** questions corresponding to each Paragraph. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If **ONLY** the correct Integer value is entered. There is **NO negative marking**.
Zero Marks: 0 In all other cases.

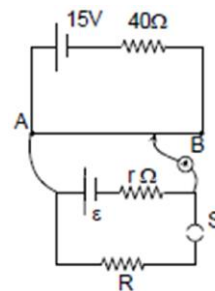
SUBJECT I : PHYSICS

60 MARKS

SECTION-1 | Type A

This section consists of 6 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. In the shown potentiometer setup, length of AB is 10 m and it has a resistance $20\ \Omega$. If the switch in the secondary circuit is open, the galvanometer gives zero deflection at 240 cm from point A . If the switch is closed and $R = x\ \Omega$ the zero deflection is at 180 cm from point A . If the switch is closed and $R = \frac{x}{3}\ \Omega$, the zero deflection is at $y\text{ cm}$ from point A . Then the value of y is:

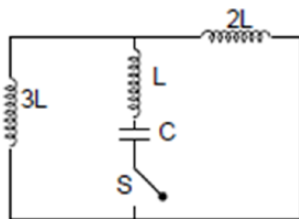


- (A) 90 (B) 120 (C) 60 (D) 80

2. A projectile is fired vertically up from the surface of earth with a velocity kV_e , where V_e is the escape velocity and $k < 1$. The maximum height upto which the projectile will rise from the surface of earth is:

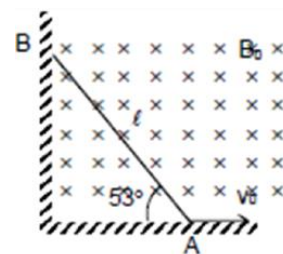
- (A) $R(k^2 + 1)$ (B) $\frac{Rk^2}{1 - k^2}$ (C) $\frac{R}{1 + k^2}$ (D) $\frac{R}{1 - k^2}$

3. Initially key is open there is no current in the inductors and capacitor is charged to a potential difference V_0 . Then the maximum energy stored in the inductor (L) after the switch is closed is:



- (A) $\frac{5}{11}CV_0^2$ (B) $\frac{5}{22}CV_0^2$ (C) $\frac{5}{33}CV_0^2$ (D) $\frac{5}{44}CV_0^2$

4. The end points of a uniform conducting rod slide on the two mutually perpendicular non-conducting surfaces in a uniform magnetic field as shown. If velocity of end A is v_0 and the rod makes an angle of 53° with the horizontal as shown. Then the maximum value of $|V_A - V_P|$ is : (where P is a point on the rod) (V_A & V_P are potential of point A and P respectively).

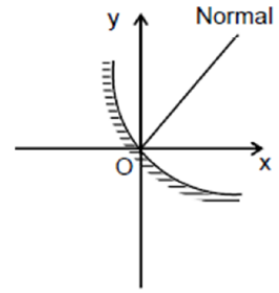


- (A) $\frac{9}{125}B_0lv_0$ (B) $\frac{18}{125}B_0lv_0$ (C) $\frac{16}{125}B_0lv_0$ (D) $\frac{32}{125}B_0lv_0$

5. In a setup of photoelectric experiment, a light of intensity 384 W/m^2 is used which is equally distributed among three wavelengths 620 nm , 496 nm and 310 nm . The light is incident perpendicularly on a surface of area 0.01 cm^2 , having work function of 2.25 eV . Assuming there is no loss of light and efficiency of emission of electrons from the surface is 50%. The photocurrent in the circuit is:

- (A) $83.2 \times 10^{-6} \text{ amp}$ (B) $166.4 \times 10^{-6} \text{ amp}$
 (C) $20.8 \times 10^{-6} \text{ amp}$ (D) $41.6 \times 10^{-6} \text{ amp}$

6. A ray is incident on a curved spherical mirror of radius of curvature 10 cm along a vector $(-\hat{i} - 2\hat{j} + 2\hat{k})$. The direction of normal at the point of incidence is along the vector $(3\hat{i} + 4\hat{j})$ as shown in the figure. The unit vector along the reflected ray is:

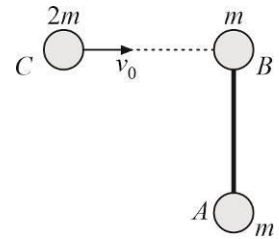


- (A) $\frac{41}{75}\hat{i} + \frac{38}{75}\hat{j} - \frac{2}{3}\hat{k}$ (B) $\frac{41}{75}\hat{i} - \frac{38}{75}\hat{j} + \frac{2}{3}\hat{k}$
 (C) $\frac{41}{75}\hat{i} + \frac{38}{75}\hat{j} + \frac{2}{3}\hat{k}$ (D) $-\frac{41}{75}\hat{i} - \frac{38}{75}\hat{j} - \frac{2}{3}\hat{k}$

SECTION-1 | Type B

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

7. Two discs A and B of mass $m = 2\text{ kg}$ each are connected by a light rod of length $l = 1\text{ m}$. These discs are placed on a smooth horizontal surface. Another disc C of mass $2m = 4\text{ kg}$ moving horizontally with velocity $v_0 = 3\text{ m/s}$ hits disc B as shown in the figure. The coefficient of restitution is $1/2$. Choose the correct option(s).



- (A) The velocity of A just after collision is zero
 (B) Loss of energy after collision between B and C is 4.5 J
 (C) Initial impulsive tension of light rod at the time of collision is zero
 (D) Angular velocity of rod just after collision is 3 rad/s

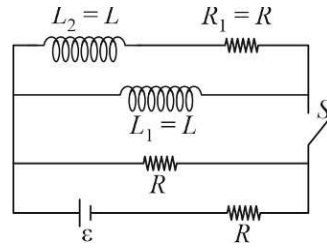
8. The acceleration $a(t)$ of a particle moving on the X-axis varies with time as:

$$a(t) = 3t - t^2 \quad \text{for } 0 \leq t < 2$$

$$a(t) = -4 + 5t - t^2 \quad \text{for } 2 \leq t < 4$$

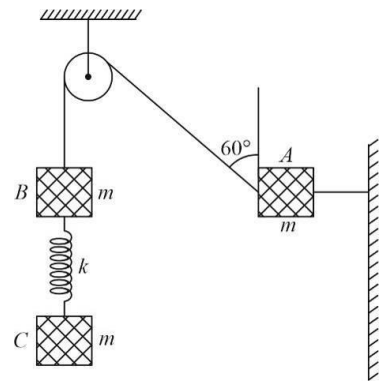
At $t = 0$, the particle is at rest at $x = 0$. Which of these options is/are correct?

- (A) The velocity of the particle at $t = 1$ is $\frac{7}{6}$
 (B) The displacement of the particle between $t = 1$ and $t = 3$ is 6.66
 (C) The velocity of the particle at $t = 3$ is $\frac{13}{2}$
 (D) The position of the particle at $t = 4$ is $x = 13.33$
9. The switch S, after being closed for a very long time is now opened. Then choose the correct options.



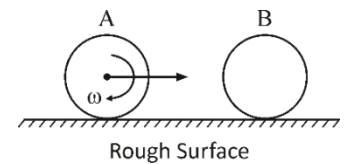
- (A) The total amount of heat generated in the resistor R_1 after that operation is $\frac{L\varepsilon^2}{2R^2}$
- (B) The total amount of heat generated in the resistor R_1 after that operation is $\frac{L\varepsilon^2}{4R^2}$
- (C) The direction of current in L_1 always remains the same
- (D) The direction of current in L_1 changes direction

10. In the adjacent figure the block A, B and C are in equilibrium. The spring of stiffness k is light and vertical. The horizontal string joining the block A to a fixed wall is cut. Choose the correct option(s), just after cutting the string.



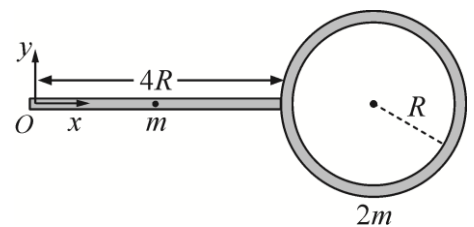
- (A) Acceleration of B is $3g/2$
- (B) Acceleration of B is $3g/4$
- (C) Acceleration of A is same as acceleration of B
- (D) Acceleration of A is more than that of B

11. A solid sphere A is rolling without slipping on a rough surface with angular speed ω as shown in figure. It collides elastically with another identical sphere B at rest. Assume no friction acts between the spheres during collision. Then : [Radius of sphere is R]



- (A) After collision A and B move in opposite directions
- (B) After collision A and B move in same direction
- (C) After collision A stops permanently and only B moves
- (D) Velocity of centre of B after it starts pure rolling on the surface is $\frac{5}{7}\omega R$

12. A uniform rod of mass m and length $4R$ is joined with a uniform ring of mass $2m$ and radius R to form a rigid body as shown. The body is in x - y plane with the origin at one end of rod as shown. Choose correct options.



- (A) Coordinates of centre of mass of the body is $(2.5R, 0)$
- (B) Coordinates of centre of mass of the body is $(4R, 0)$
- (C) Moment of inertia of the body about an axis passing through its centre of mass and perpendicular to x - y plane is $\frac{28}{3}mR^2$
- (D) Moment of inertia of the body about z -axis is $48mR^2$

SECTION-2

This section consists of THREE (03) Paragraphs. There are **TWO (02)** questions corresponding to each Paragraph. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08).

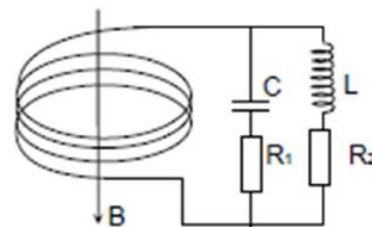
Paragraph for Question 1-2

Two small positively charged spheres are suspended from a common point at the ceiling by the insulating light strings of equal length. The first sphere has mass m_1 and charge q_1 while the second one has mass m_2 and charge q_2 . At equilibrium, the first string makes angle θ_1 with the vertical and the second string makes an angle θ_2 with the vertical.

- Find the value of θ_1 (in degrees) if $m_1 = m_2$, $q_1 = 2q_2$ and $\theta_2 = 30^\circ$.
- Find the value of θ_1 (in degrees) if $m_2 = \sqrt{2}m_1$, $q_1 = 13q_2$ and $\theta_2 = 45^\circ$.

Paragraph for Question 3-4

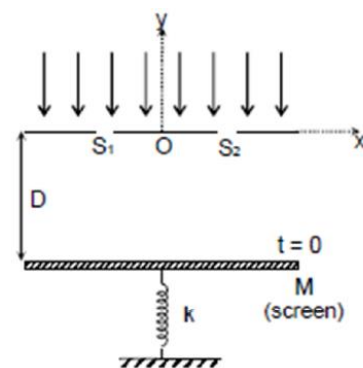
Consider an electric circuit consisting of a coil of negligibly small inductance, consisting of $N = 10$ turns and with the surface area of a single loop $S = 10 \text{ cm}^2$, resistors $R_1 = R_2 = 3\Omega$, capacitor $C = 2/9 \text{ F}$ and an inductance $L = 1\text{H}$, connected as shown in the figure. At the moment of time $t = 0$ a magnetic field, parallel to the axis of the coil is switched on. The induction of the magnetic field starts growing linearly, starting from $B = 0$ until the maximal value $B = 1\text{T}$ is achieved at $t = 10 \text{ ms}$. Further the induction of the magnetic field remains constant (and equal to 1T). Given $e^{-R_2/L \times 5\text{ms}} = 0.985$



- Find the current through the resistors R_2 (in milli ampere) at the moment of time $t_1 = 5 \text{ ms}$.
- What is the maximum value of charge (in milli coulomb) that gets deposited on capacitor plate.

Paragraph for Question 5-6

A monochromatic light of wavelength λ is incident on two slits S_1 and S_2 made on x-axis. The mid point of the line S_1S_2 is considered as origin and 'd' is the distance between the two slits which are symmetric about y-axis. The interference pattern is observed on a plate of mass M which is acting as a screen. The screen is attached to one end of a vertical spring of spring constant 'k' and the other end of spring is attached with the ground. Spring is massless and screen always remains in horizontal plane. The distance between the plane of slits and the screen is D ($D \gg d$). Assuming S_1 and S_2 are coherent in nature. At $t = 0$, screen is released and spring is in its natural length. It is given that $M = 1\text{kg}$,



$k = 1 \text{ N/m}$, $D = \frac{5Mg}{k}$, $g = 10 \text{ m/s}^2$, $d = 200\lambda$. The x co-ordinate

(in cm) of the first order maxima on the screen as a function of time is given by $\pm(\lambda_1 - \lambda_2 \cos t)$.

5. The value of λ_1 is _____.

6. The value of λ_2 is _____.

SPACE FOR ROUGH WORK

SECTION-1 | Type A

This section consists of 6 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- Which of the following statement is correct?
 - In fluorite structure anions form FCC lattice and cations occupy all tetrahedral voids.
 - If the radius of cation and anion are $0.214 \text{ \AA}$ & $0.95 \text{ \AA}$ respectively then co-ordination number of cation in the crystal is 4
 - An atom/ion is transferred from a lattice site to an interstitial position in Schottky defect
 - Density of crystal always increases due to substitutional impurity defect.
- Ammonium compound which on heating does not give NH_3 is:
 - $(\text{NH}_4)_2\text{SO}_4$
 - $(\text{NH}_4)_2\text{CO}_3$
 - NH_4NO_2
 - NH_4Cl
- 11 moles N_2 and 12 moles of H_2 mixture reacted in 20 litre vessel at 800 K. After equilibrium was reached, 6 mole of H_2 was present. 3.58 litre of liquid water is injected in equilibrium mixture and resultant gaseous mixture suddenly cooled to 300 K. What is the final pressure of gaseous mixture? Neglect vapour pressure of liquid solution. Assume (i) all NH_3 dissolved in water (ii) no change in volume of liquid (iii) no reaction of N_2 and H_2 at 300 K:
 - 18.47 atm
 - 60 atm
 - 22.5 atm
 - 45 atm
- The incorrect statement about the following reaction sequence is:

$\text{Benzene} \xrightarrow{(1)} \text{P} \xrightarrow{(2)} \text{Q} \xrightarrow{(3)} \text{R} \xrightarrow{(4)} \text{S} \xrightarrow{(5)} \text{Product}$

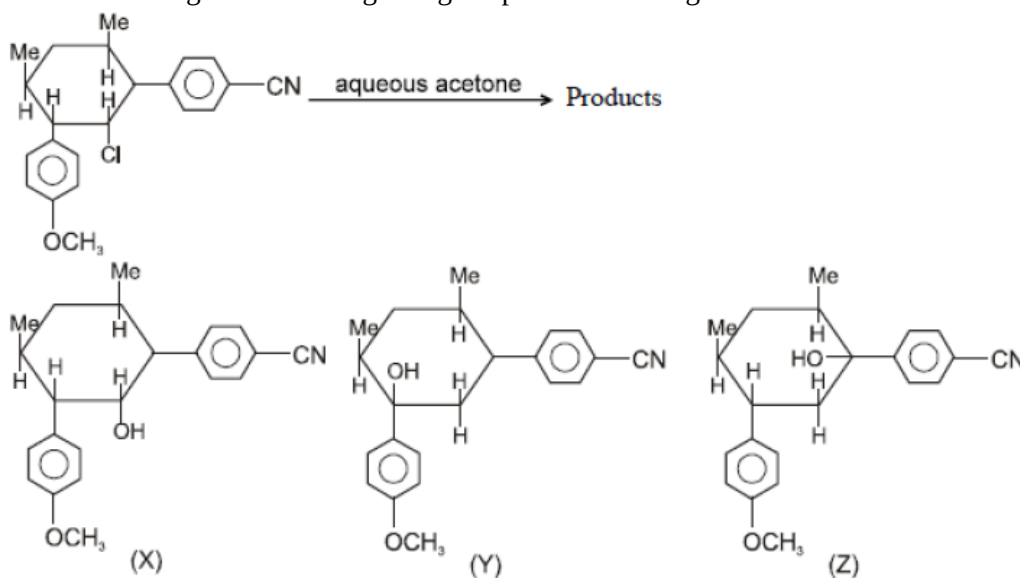
 - T gives an aldol condensation reaction on heating with NaOH solution
 - Compound R is more reactive towards N_2H_4 than compound Q
 - Compound Q $\xrightarrow[\text{0-5}^\circ\text{C}]{\text{(i) Sn/HCl, (ii) NaNO}_2\text{-HCl}}$ X $\xrightarrow{\text{H}_3\text{PO}_2}$ P; P is chlorobenzene
 - In step 5 Acetaldehyde is more reactive than T
- 20 ml $\frac{\text{M}}{10} \text{H}_3\text{PO}_4$ solution is treated with 40 ml of $\frac{\text{M}}{10} \text{NaOH}$ solution pK_{a_1} , pK_{a_2} and pK_{a_3} of H_3PO_4 are 2.15, 7.2 and 12.3 respectively. The pH of the resulting solution is:
 - 7.35
 - 7
 - 4.675
 - 9.75

6. Compound A, $C_5H_{10}O_5$ gives a tetraacetate with $(CH_3CO)_2O$ and oxidation of A with $Br_2 - H_2O$ gives an acid, $C_5H_{10}O_6$. Reduction of A with red P and HI gives iso-pentane, there are two possible structures of compound A which can be distinguished by using :
- (A) HIO_4 (B) Red P + HI
(C) Phenyl hydrazine (D) Tollen's reagent

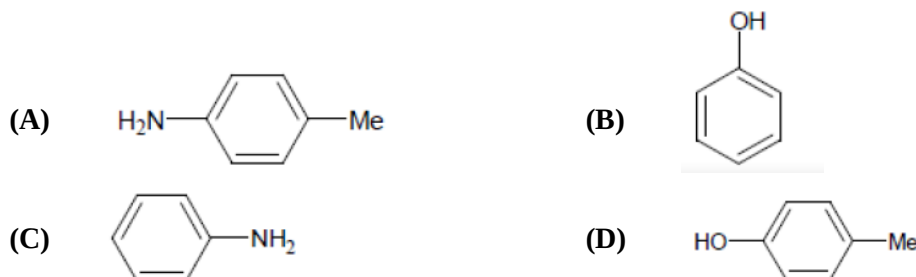
SECTION-1 | Type B

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

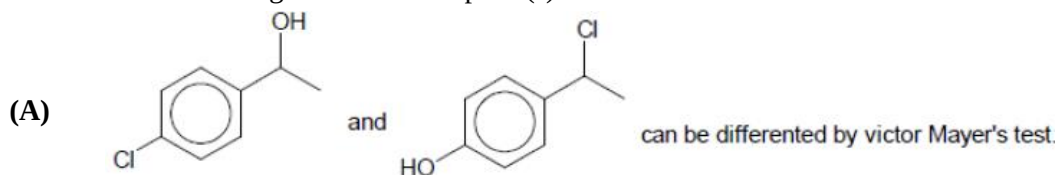
7. Which of the following are correct regarding the products of the given reaction?



- (A) only X and Z are formed (B) $Y > Z$ (amount)
(C) $X > Y$ (amount) (D) $Y > X$ (amount)
8. p-Toluidine reacts with benzenediazonium chloride to form a compound, which on boiling with H_2SO_4 , gives product/s:



9. Which of the following is/are correct option(s):



- (B) Monomer of orlon is acrylonitrile
 (C) Aspartame is the methyl ester of dipeptide of L-aspartic acid and L-phenylalanine
 (D) D-Glucose, D-Fructose and L-Mannose give same osazone with $\text{Ph}-\text{NH}-\text{NH}_2$

10. Which of the following reaction(s) produce(s) Cr_2O_3 along with O_2 ?

- (A) Heating $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ (B) Heating CrO_3
 (C) Heating $\text{K}_2\text{Cr}_2\text{O}_7$ (D) Reaction of Al_2O_3 with Cr

11. The total vapour pressure of a binary solution is given by :

$$P = (80 X_A + 240 X_B) \text{ mm Hg}$$

Where X_A and X_B are the mole fractions of components A and B in solution. Choose the correct statement(s) from the following :

- (A) Vapour pressure of solution is less than that of the pure B component
 (B) Vapour pressure of solution is more than that of pure A component
 (C) Vapour pressure of pure A is 80 mm of Hg and that of pure B is 240 mm Hg
 (D) The vapour pressure of pure A and B are 240 and 80 mm Hg respectively

12. The correct statement(s) pertaining to colloids is/are:

- (A) The process of converting freshly prepared precipitate into colloidal sol by shaking it with dispersion medium in the presence of excess amount of electrolyte is called peptization
 (B) Lyophilic colloids are also called as reversible sols
 (C) Sulphur sol is an example of multi-molecular sols
 (D) Sols are those colloids in which dispersion medium is liquid phase while dispersed phase is solid phase

SPACE FOR ROUGH WORK

SECTION-2

This section consists of THREE (03) Paragraphs. There are TWO (02) questions corresponding to each Paragraph. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08).

Paragraph for Question 1-2

In the manufacture of $\text{Na}_2\text{CO}_3(\text{s})$ by Solvay process, $\text{NaHCO}_3(\text{s})$ is decomposed by heating:



$$K_p = 0.23 \text{ at } 100^\circ\text{C} \quad \Delta H^\circ = 136 \text{ kJ}$$

1. If a sample of $\text{NaHCO}_3(\text{s})$ is brought to a temperature of 100°C in a closed container total gas pressure (in atm) at equilibrium is:
2. A maximum of 1.00 mol each of $\text{NaHCO}_3(\text{s})$ and $\text{Na}_2\text{CO}_3(\text{s})$ is introduced into a 2.5 L flask in which $P_{\text{CO}_2} = 2.10 \text{ atm}$ and $P_{\text{H}_2\text{O}} = 0.94 \text{ atm}$. When equilibrium is established at 100°C , then partial pressure (in atm) of CO_2 :

Paragraph for Question 3-4

The process of selectively precipitating an ion from a solution of more than one ion is called selective precipitation. The selective precipitation of ions from a solution in the form of salt can be done by adding precipitating agent slowly. AgNO_3 solid is slowly added to one litre of a solution (without changing the volume) containing 0.1 mole of Cl^- and 0.1 mole of PO_4^{3-} . $K_{\text{sp}}\text{AgCl} = 1.2 \times 10^{-10}$ and $K_{\text{sp}}\text{Ag}_3\text{PO}_4 = 2.7 \times 10^{-18}$.

Answer the following questions based on the above paragraph.

3. Minimum concentration of Ag^+ required for the precipitation of Ag_3PO_4 is $a \times 10^{-b} \text{ M}$, value of $(a + b)$ is:
4. The concentration of Cl^- in the solution when Ag_3PO_4 starts precipitating is $z \times 10^{-w} \text{ M}$, then $\frac{z \times w}{4}$ is:

Paragraph for Question 5-6

Stereoisomeric complexes have same molecular formula, same constitution they differ only with respect to the spatial orientation of ligand in space around the metal ion.

5. The possible number of geometrical isomers for the complex $[\text{Pt}(\text{bn})_2]^{2+}$ are [where bn is $\text{H}_3\text{C}-\text{CH}(\text{NH}_2)-\text{CH}(\text{NH}_2)-\text{CH}_3$].
6. The possible number of stereoisomers for the same complex $[\text{Pt}(\text{bn})_2]^{2+}$ are.

SPACE FOR ROUGH WORK

SECTION-1 | Type A

This section consists of 6 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Let $f(x) = \lim_{h \rightarrow 0} \left(\frac{(\sin(x+h))^{\ln(x+h)} - (\sin x)^{\ln x}}{h} \right)$, then $f\left(\frac{\pi}{2}\right)$ is:
- (A) 0 (B) 1 (C) $\ln\left(\frac{\pi}{2}\right)$ (D) None of these
2. Let $I = \int_0^1 \frac{\sin x}{\sqrt{x}} dx$ and $J = \int_0^1 \frac{\cos x}{\sqrt{x}} dx$. Then which one of the following is true?
- (A) $I > \frac{2}{3}$ and $J > 2$ (B) $I < \frac{2}{3}$ and $J < 2$
- (C) $I < \frac{2}{3}$ and $J > 2$ (D) $I > \frac{2}{3}$ and $J < 2$
3. The function $f(x) = \begin{cases} \frac{a|x^2 - x - 6|}{6 + x - x^2}, & x < 3 \\ b \lim_{x \rightarrow 3} \left(\frac{x \int_0^x t dt}{x - 3} \right), & x = 3 \\ \frac{x - [x]}{x - 3}, & x > 3 \end{cases}$ is continuous at $x = 3$. If $f(3) \neq 0$, then (a, b) is
- (where $[\cdot]$ denotes the greatest integer function).
- (A) $(3, 1)$ (B) $\left(\frac{1}{3}, 1\right)$ (C) $\left(1, \frac{1}{3}\right)$ (D) $\left(1, \frac{1}{9}\right)$
4. Let $g(x) = f_n(x)$, where $f_2(x) = f(f(x))$, $f_3(x) = f(f(f(x))) \dots$ and $f(x) = x^3 - \frac{3}{2}x^2 + x + \frac{1}{4}$ then $\int_{1/4}^{3/4} g(x) dx$ is:
- (A) $\frac{n}{2} + 2$ (B) $2^n + 1$ (C) $\frac{1}{4}$ (D) $\frac{3}{4}$

5. Let $\vec{r} = x\vec{a} + y\vec{b}$ where x, y are constant and $\vec{r}$ is a fixed vector such that $\frac{|\vec{r}|^2}{xy} \leq 4\vec{a} \cdot \vec{b}$, then $\frac{|\vec{a}|}{|\vec{b}|}$ is equal to (where $xy > 0$).
- (A) 1 (B) $\frac{x^2}{y^2}$ (C) $\frac{x}{y}$ (D) $\frac{y}{x}$
6. The vectors $(x^2 - 1)\hat{i} + 2(x^2 - 1)\hat{j} - 3(x^2 - 1)\hat{k}$, $(2x^2 - 1)\hat{i} + (2x^2 + 1)\hat{j} + x^2\hat{k}$, and $(3x^2 + 2)\hat{i} + (x^2 + 4)\hat{j} + (x^2 + 1)\hat{k}$ are non-coplanar. The number of real values that x cannot take is:
- (A) 0 (B) 1 (C) 2 (D) 4

SECTION-1 | Type B

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

7. Which of the following can be antiderivative of $f(x) = \frac{x^4 + 1}{x^6 + 1}$? (Where 'C' is the constant of integration).
- (A) $\tan^{-1}x + \frac{1}{3}\tan^{-1}(x^3) + C$ (B) $\tan^{-1}\left(x - \frac{1}{x}\right) - \frac{2}{3}\tan^{-1}(x^3) + C$
- (C) $\frac{1}{3}\tan^{-1}x + \tan^{-1}(x^3) + C$ (D) $\frac{2}{3}\tan^{-1}\left(x - \frac{1}{x}\right) - \tan^{-1}(x^3) + C$
8. Consider a system of linear equations in three variables.
- $$\begin{aligned} -x + y + z &= a \\ x - y + z &= b \\ x + y - z &= c \end{aligned}$$
- Then:
- (A) $x + y = \frac{a+b}{2} + c$ (B) $y + z = a + \frac{b+c}{2}$
- (C) $x + z = \frac{a+c}{2} + b$ (D) $x + y + z = a + b + c$
9. Let T be the Δ with vertices $(0, 0)$, $(0, c^2)$ and (c, c^2) , $c > 0$ and let R be the region between $y = cx$ and $y = x^2$ where $x > 0$ then:
- (A) Area of $(R) = \frac{c^3}{6}$ (B) Area of $R = \frac{c^2}{3}$
- (C) $\lim_{c \rightarrow 0^+} \frac{\text{Area}(T)}{\text{Area}(R)} = 3$ (D) $\lim_{c \rightarrow 0^+} \frac{\text{Area}(T)}{\text{Area}(R)} = \frac{3}{2}$

10. Let $\vec{a}, \vec{b}, \vec{c}$ be three coplanar unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. If three vectors $\vec{p}, \vec{q}, \vec{r}$ parallel to $\vec{a}, \vec{b}, \vec{c}$ respectively and having integral but different magnitudes, the among the following options $|\vec{p} + \vec{q} + \vec{r}|$ can take a value equal to:
 (A) 1 (B) 0 (C) $\sqrt{3}$ (D) 2
11. Two players A and B toss a coin which has a probability p of showing head ($0 < p < 1$), in the cyclic order A, A, B, A, A, B, ... till a head appears. Let x denotes the probability that A gets the head first and y that of B, then:
 (A) $x + y = 1$ (B) $x > y$ for all p (C) $x > \frac{1}{2}$ for all p (D) $y > \frac{1}{2}$ for all p
12. Let a, b, c be the sides of a triangle whose perimeter is P and area is A , then:
 (A) $P^3 < 27(b+c-a)(c+a-b)(a+b-c)$
 (B) $P^2 \leq 3(a^2 + b^2 + c^2)$
 (C) $a^2 + b^2 + c^2 \geq 4\sqrt{3}A$
 (D) $P^4 < 256A^2$

SECTION-2

This section consists of THREE (03) Paragraphs. There are **TWO (02)** questions corresponding to each Paragraph. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08).

Paragraph for Question 1-2

Vector $\vec{a}, \vec{b}, \vec{c}$ are three non-zero vectors such that $(\vec{a} - \vec{b}) \cdot \vec{c} = 0$.

Let $\vec{a} \times (\vec{b} \times \vec{c}) + \vec{b} \times (\vec{c} \times \vec{a}) = (4 + x^2)\vec{b} - (4x \cos^2 \theta)\vec{a}$, where $\vec{a}$ and $\vec{b}$ are non-collinear vectors and $x > 0, 0 < \theta < 10$, then:

- Number of different value(s) of x will be:
- Number of different value(s) of θ will be:

Paragraph for Question 3-4

Read the following write up carefully and answer the following question:

Let $y = f(x)$ be a curve such that $x = 1 - 3t^2, y = t - 3t^3$, where t is a parameter and $f(x)$ is not identically zero $\forall t \in R$. The tangent drawn at $P(t)$ to the curve makes an angle θ with positive direction of x -axis.

3. If the tangent at $A \equiv (-2, 2)$ meet the curve again at B then coordinates of point B is $\left(-\frac{p}{3}, -\frac{q}{9}\right)$. Find out $p + q$ _____.
4. The acute angle between the tangents at $x = 0$ to the curve $y = f(x)$ is $\frac{k\pi}{9}$, then $k =$ _____.

Paragraph for Question 5-6

Let $f(x) = 12x^2 \int_0^1 y f(y) dy + 20 \int_0^1 xy^2 f(y) dy + 4x$

5. If maximum value of $f(x)$ is $\frac{\lambda}{8}$, then $\lambda =$ _____.
6. Number of solutions of equation $|f(|x|)| = e^{|x|}$ is:

SPACE FOR ROUGH WORK